CNOT costs of controlled balanced involutions and an invariant two-SQiSW criterion

Gregory Morse Peter Rakyta

7 November 2026

Abstract

We classify the ancilla-free CNOT cost of controlling a traceless Hermitian involution on two qubits, allowing arbitrary one-qubit unitaries. The cost is four at operator-Schmidt rank four, three at rank two, and one or two for rank-one operators according to their local support. With unrestricted two-qubit gates these costs are respectively three, three, one, and two. The rank-four lower bound holds for arbitrary Ising-class interactions, including variable interaction angles. Controlled fermionic SWAP illustrates the distinction: a literal four-CNOT, six-rotation circuit attains its Ising minimum, while three unrestricted interactions suffice. We also express the known two-SQiSW Weyl-chamber region as a necessary-and-sufficient test on scalar-independent polynomial invariants, avoiding Weyl canonicalization when deciding membership. Exact symbolic constructions, explicit phase contracts, and row-resolved certificates accompany these results in a reference library. Comparisons with Qiskit, tket, BQSKit, Quartz, and QUESO separate construction quality from lower bounds; in particular, BQSKit also finds four-CNOT, six-local-gate circuits. The reference grid and Clifford atlas are reproducibility resources, rather than evidence of all-grid optimality or compiler dominance.

1 Introduction and related work

Controlling an operator preserves more structure than its ordinary two-qubit local-equivalence class. In particular, local factors in a target block become conditional operations. We exploit this observation to classify a continuous class of three-qubit gates: $C(V) = \mathrm{diag}(I_4, V)$, where $V = V^\dagger$, $V^2 = I_4$, and $\mathrm{tr}\, V = 0$. A two-angle normal form gives matching constructions on every operator-Schmidt-rank stratum. A separate control-space argument excludes three Ising interactions on the rank-four stratum. The resulting classification is the main contribution.

Gate-count lower bounds based on operator-Schmidt strength originate in work including Nielsen et al. [11]. Yu, Duan and Ying proved that Toffoli requires five unrestricted two-qubit gates [12]; Yu and Ying classified controlled-controlled gates and proved the five-gate unrestricted Fredkin bound [13]. Their controlled-product arguments are directly relevant here. Shende and Markov studied block-diagonal three-qubit circuits with fewer than six CNOTs and proved Toffoli's six-CNOT minimum [14]. Our Ising proof uses the same general principle of restricting intermediate control frames, with an explicit Schmidt-space rigidity lemma. Balanced involutions exclude the nonbalanced relative blocks of ordinary Fredkin and Toffoli; our smaller counts concern a different class.

Two-qubit invariant and Cartan methods are established [1, 2]. The universal two-B construction is due to Zhang et al. [3]. Huang et al. give analytical SQiSW synthesis, its two-pulse Weyl region, and its universal three-pulse bound [4]. Our second result is an invariant reformulation of that known region, with an exact algebraic proof and implementation. Our symbolic charts implement

these established pulse counts with scalar tracking and explicit degeneracy branches; the counts and the existence of analytical local frames are prior results.

The accompanying software joins OSR factorization, native analytical charts, and exact local rewrites. Quartz [5] and QUESO [6] already synthesize verified, gate-set-dependent optimizations. UniversalQCompiler provides analytical synthesis in Mathematica [7]. Single-qubit exact Clifford+T synthesis and normal forms have an extensive literature [15, 16, 17, 18]; the 192 phase-resolved one-qubit Clifford circuits in our atlas are a small reference and regression suite. Neither their number nor an easily reproducible shortest-path search is presented as a major synthesis advance. Earlier numerical OSR partition resynthesis [10] is a separate project and supplies no numerical evidence for the identities proved here.

2 Resources, phase, and operator-Schmidt rank

All gate-count theorems use arbitrary one-qubit unitaries, all-to-all connectivity, fixed input/output wires, arbitrary input states, and no ancillas. An Ising interaction is locally equivalent to e^{itZZ}; its angle may vary between uses. CNOT is one member of this class. An unrestricted two-qubit gate is a different, strictly larger resource. For supplied hardware alphabets, the archive counts the declared primitive; a theorem in the relaxed model transfers as a lower bound, and requires a separate native construction to prove a matching optimum.

Literal synthesis requires $U = C$. Projective synthesis requires

$$U = e^{i\gamma} C, \qquad |e^{i\gamma}| = 1, \tag{1}$$

with the scalar recorded at zero physical cost. A physical GP gate is never inserted. At equal declared cost we prefer literal equality. Controlling a projective word requires $P(\gamma)$ on the added control, since $C(U) = P_c(\gamma)C(C)$. The release proves a concrete controlled-Sycamore substitution both with and without this correction. This illustrates the contract; SDKs that track global phase can already implement this rule.

For a cut $A|B$, realign $U_{ab,a'b'}$ with row (a, a') and column (b, b'). The rank is $r_{A|B}(U) = \mathrm{OSR}_{A|B}(U)$. It obeys

$$r_{A|B}(UV) \le r_{A|B}(U) r_{A|B}(V), \qquad r_{A|B}(L_A \otimes L_B) = 1. \tag{2}$$

For a controlled operator, its *control OSR* means this rank on $c|ab$. If its two target branches are linearly independent it is two, and its control-side Schmidt space is the full diagonal space $\mathcal{D} = \mathrm{span}\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$. For a target cut, the V block is a submatrix of the realignment of $C(V)$. Thus $\mathrm{OSR}(V) = 4$ forces ranks $(2, 4, 4)$ on the three singleton cuts. Every crossing Ising interaction has rank at most two, so Eq. (2) constrains the number and placement of interactions.

3 A rank-four controlled-block lower bound

Theorem 1 *If* $\mathrm{OSR}(V) = 4$*, then* $C(V)$ *requires at least four Ising-class interactions, even with arbitrary interaction angles, arbitrary one-qubit unitaries, and an arbitrary scalar. Controlled fermionic SWAP attains this bound with four CNOTs.*

3.1 Control-space rigidity

Lemma 1 *If invertible one-qubit matrices* P, Q *map the full diagonal matrix space*

$$\mathcal{D} = \mathrm{span}\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$$

onto itself by $D \mapsto PDQ$, then P, Q are monomial with the corresponding common permutation. Thus locally equivalent controlled operators with control OSR two have corresponding conditional target blocks, up to phases, exchange of the two branches, and common target-local input/output frames.

Proof. The only rank-one lines in $\mathcal{D}$ are its two coordinate projectors: $\det \operatorname{diag}(a, b) = ab$. An invertible left/right map preserves rank and must permute these two lines. Its column and row factors therefore have the stated form. Control Schmidt spaces of rank-two controlled operators are full copies of $\mathcal{D}$, proving the second claim. In particular, a locally equivalent realization of $\operatorname{diag}(\mathbb{I}, V)$ has a local conditional branch. Its relative target block has the same OSR as V, also after branch inversion. *QED*

3.2 Exclusion of three interactions

Proof of the lower bound in Theorem 1. Call the targets a, b. The singleton cuts have ranks $2, 4, 4$ for c, a, b: the rank-four assertion follows from the V block, since a target cut has maximum rank four. Each crossing Ising interaction has rank at most two. Consequently a and b each have degree at least two and c at least one in an interaction graph. Two interactions cannot satisfy these demands. With three, the only edge multisets are a triangle ca, cb, ab, or two ab edges and one control edge. The triangle has six time orders. The doubled-target-edge case has three placements of the control edge for each of its two possible targets, also six. These are the twelve surviving edge words.

First consider one control crossing and two internal target interactions. Absorb local frames into adjacent gates and put the single crossing in controlled form. Its target branches are W_+W_- and W_+QW_-, where Q is a one-wire unitary and $W_\pm$ contain the target operations after/before the crossing. By Lemma 1, choose a local branch as the first one (inverting Q if the branches are exchanged). If both internal interactions lie on one side, the corresponding product must be local, so the relative branch is local. If one lies on each side, the relative branch is $W_-^\dagger QW_-$ with W_- locally Ising. To make the conjugation bound explicit, write the one-wire operator as $q_0 I + q_x X + q_y Y + q_z Z$. Conjugating by e^{itZZ} leaves I and Z unchanged, sends $X \otimes I$ to a combination of $X \otimes I$ and $Y \otimes Z$, and sends $Y \otimes I$ to a combination of $Y \otimes I$ and $X \otimes Z$. Consequently the result is $M_0 \otimes I + M_1 \otimes Z$ and has OSR at most two. Outer local frames preserve that rank. In neither case can the relative block have rank four.

Now consider the triangle. Put each control interaction in controlled form and combine all operations on the control between them into a one-qubit unitary N. Local frames at the endpoints are harmless by Lemma 1. A 2×2 unitary is either monomial or has all four entries nonzero. If N is monomial, the circuit remains controlled after a possible control permutation. Each conditional target branch contains the sole internal ab interaction, with only local factors around it. That interaction must be entangling; otherwise the singleton rank demands already fail. Neither branch is local, a contradiction.

If N has all entries nonzero, factor the target operation K between the crossings to one side. The four control blocks then span $\{\mathbb{I}, B, A, AB\}$, where B is a nontrivial one-wire unitary on one target and A is the other controlled target action conjugated through K. Their span dimension equals the control OSR, namely two. If the internal interaction is outside the crossings, K is local, and A, B act nontrivially on different wires. The four tensor products are independent, contradicting rank two. If the internal interaction lies between them, independence of $\mathbb{I}, B$ forces $A \in \operatorname{span}\{\mathbb{I}, B\}$. The transformed controlled gates then act only on c, b. Apart from local factors the whole circuit is $(\mathbb{I}_c \otimes K_{ab}) W_{cb}$. Its $a|cb$ rank is at most $\mathrm{OSR}(K) = 2$, again a contradiction. These cases exclude every three-interaction word. The four-CNOT identity in Sec. 4.3 attains the bound for U_F. *QED*

The supplied checker verifies the exact matrix identity, cut minors, three-edge enumeration, Ising conjugation identities, and native-class identifications. The argument above is an analytical proof with checked algebraic components, not a proof-assistant formalization or a conclusion from failed numerical optimization.

4 Complete classification for balanced involutions

Theorem 2 *Let $V = V^\dagger$, $V^2 = I_4$, and $\operatorname{tr} V = 0$. The minimum CNOT and unrestricted two-qubit-gate counts of $C(V)$ are:*

Relative block	*CNOT*	*Unrestricted*
$\mathrm{OSR}(V) = 4$	*4*	*3*
$\mathrm{OSR}(V) = 2$	*3*	*3*
$\mathrm{OSR}(V) = 1$, *one-body Pauli up to local conjugacy*	*1*	*1*
$\mathrm{OSR}(V) = 1$, *two-body Pauli product up to local conjugacy*	*2*	*2*

There is no rank-three stratum. Multiplying V by an arbitrary scalar phase does not change these interaction counts when arbitrary $U(2)$ locals are available: the scalar is supplied by a physical one-qubit phase on c.

4.1 Two-angle normal form, including degeneracies

Expand a traceless Hermitian V in the real Pauli basis:

$$V = \vec{a} \cdot \vec{\sigma} \otimes I + I \otimes \vec{b} \cdot \vec{\sigma} + \sum_{ij} T_{ij} \sigma_i \otimes \sigma_j. \tag{3}$$

Multiplying Pauli matrices and setting $V^2 = I$ gives exactly

$$|a|^2 + |b|^2 + \|T\|_F^2 = 1, \quad Tb = 0, \quad T^T a = 0, \quad \operatorname{cof}(T) = ab^T. \tag{4}$$

The scalar, one-body, and two-body coefficients of V^2 independently verify these equations. The supplied checker expands every coefficient.

If a, b are nonzero, rotate them to the X and Z axes respectively. Then the X row and Z column of T vanish. Its remaining real 2×2 block has a signed singular-value decomposition under the remaining rotations in the two perpendicular planes. Signs are retained, so only proper rotations, implemented by local unitaries, are needed. This gives

$$V_0 = AX_a + BZ_b + CY_aY_b + DZ_aX_b, \quad A^2 + B^2 + C^2 + D^2 = 1, \quad AB + CD = 0. \tag{5}$$

For completeness, if only a is nonzero, $\operatorname{cof}(T) = 0$ forces $\operatorname{rank} T \leq 1$ and $T^T a = 0$ makes its nonzero left singular axis perpendicular to a. The two remaining terms are put into Eq. (5) by local rotations. The case with only b is symmetric. If both vanish, T has rank one and unit norm, giving a Pauli product. If $T = 0$, exactly one of a, b is nonzero and has unit norm. Thus the same normal form covers every degenerate case.

The matrix $\left(\begin{smallmatrix} A & -C \\ D & B \end{smallmatrix}\right)$ has rank one and Frobenius norm one, so it is an outer product of two real unit vectors. Write these as $(\sin\alpha, \cos\alpha)^T$ and $(\sin\beta, \cos\beta)$. With $K = \mathrm{CX}_{a,b}$ and $Q(t) = R_y(t) Z R_y(-t)$ this proves

$$V_0 = K \left[Q(\alpha)_a Q(\beta)_b \right] K, \tag{6}$$

with coefficients $(A, B, C, D) = (\sin\alpha \sin\beta, \cos\alpha \cos\beta, -\sin\alpha \cos\beta, \cos\alpha \sin\beta)$. In the Pauli coefficient matrix the four coefficients occupy different rows and columns. Hence OSR is their number of nonzero entries, which is one, two, or four. The relation $AB + CD = 0$ excludes three.

4.2 Matching constructions and two-gate exclusions

Controlling Eq. (6) uses two outer K gates and one controlled Q on each target. Since every Q is a one-qubit Pauli axis, its controlled gate is locally CZ and costs one CNOT. This proves the four-CNOT upper bound throughout the class. Local conjugating frames for V add only one-qubit gates to $C(V)$.

On rank-two boundaries there is a stronger form $V_0 = K'Q_jK'^{\dagger}$ with K' containing one CNOT. Representative edges are shown below; $S = \mathrm{diag}(1, i)$ and t is the remaining angle.

Edge	K'	Q_j
$\alpha = 0$	$H_b\mathrm{CX}_{a,b}$	$\cos tX_b + \sin tZ_b$
$\alpha = \pi/2$	$H_aS_b\mathrm{CX}_{a,b}$	$\sin tZ_a + \cos tY_a$
$\beta = 0$	$S_a^{\dagger}\mathrm{CX}_{b,a}$	$\cos tZ_b + \sin tY_b$
$\beta = \pi/2$	$H_a\mathrm{CX}_{a,b}$	$\sin tZ_a + \cos tX_a$

Other edges follow by angle periodicity and local Pauli conjugations. Controlling Q_j between $K'^{\dagger}$ and K' costs three CNOTs. The checker proves all four identities with t free, including the corners.

To exclude two unrestricted interactions for nonlocal V, their edges must cover all three wires. If both edges meet c, the controlled-product argument of Yu and Ying [13, Proposition 2] says that the conditional branches are target-local; their relative product cannot be nonlocal. If the edges are ca, ab, there is a single crossing of the control cut. Its OSR must be two. A two-qubit unitary of OSR two is locally controlled: in its Cartan expansion, rank two leaves only one nonzero interaction coordinate. Put this gate into controlled form. With the ab interaction entirely on one side of the crossing, the local-branch argument of Lemma 1 makes the relative target branch local. The reversed order and exchanged targets give the same conclusion. Two gates therefore cannot implement any nonlocal relative V.

For any balanced V, its $+1$ and -1 eigenspaces both have dimension two. An unrestricted two-qubit change of basis J satisfies $V = JZ_aJ^{\dagger}$. The circuit $J^{\dagger}, CZ_{ca}, J$, in time order, uses three unrestricted interactions. Together with the preceding exclusion and Theorem 1, this proves both nonlocal rows of Theorem 2. In the rank-one cases, local Hermitian unitary factors reduce to identities or Pauli axes. One-body control costs one CX; a two-body Pauli product costs two. One gate cannot cover its three nontrivial singleton cuts, proving the latter minimum.

The theorem depends on more than Weyl class. The relative blocks I, Z_a, and Z_aZ_b all have the local Weyl class, whereas their controlled costs are zero, one, and two. A classification of arbitrary controlled-V by the Weyl class of V alone is therefore impossible. The present theorem classifies balanced involutions, not all controlled unitaries.

4.3 Controlled fermionic SWAP

Let $F|00\rangle = |00\rangle$, $F|01\rangle = |10\rangle$, $F|10\rangle = |01\rangle$, and $F|11\rangle = -|11\rangle$. Then $F = (Z_a + Z_b + X_aX_b + Y_aY_b)/2$, so its four independent Pauli Schmidt coefficients give $\mathrm{OSR}(F) = 4$. Equivalently, $F = (S^{\dagger} \otimes S^{\dagger})\mathrm{iSWAP}$. It is Hermitian, squares to identity, and has trace zero. Ordinary SWAP has trace two and belongs to a different class.

Figure 1 is a literal identity with four CNOTs and six R_y rotations. In time order its rotations and interactions are

$$R_y(-\pi/2)_a,\ \mathrm{CX}_{a,b},\ R_y(-\pi/4)_a,\ R_y(-3\pi/4)_b,$$
$$\mathrm{CX}_{c,a},\ \mathrm{CX}_{c,b},\ R_y(\pi/4)_a,\ R_y(3\pi/4)_b,\ \mathrm{CX}_{a,b},\ R_y(\pi/2)_a.$$

Exact multiplication gives $C(F)$ with no leftover scalar. Six is a local-count upper bound, not a proved local minimum. An independently constructed two-qubit J, rotating the odd-parity subspace by a real $\pi/4$ rotation, also gives $F = J^\dagger Z_a J$; its exact three-interaction controlled circuit is checked separately.

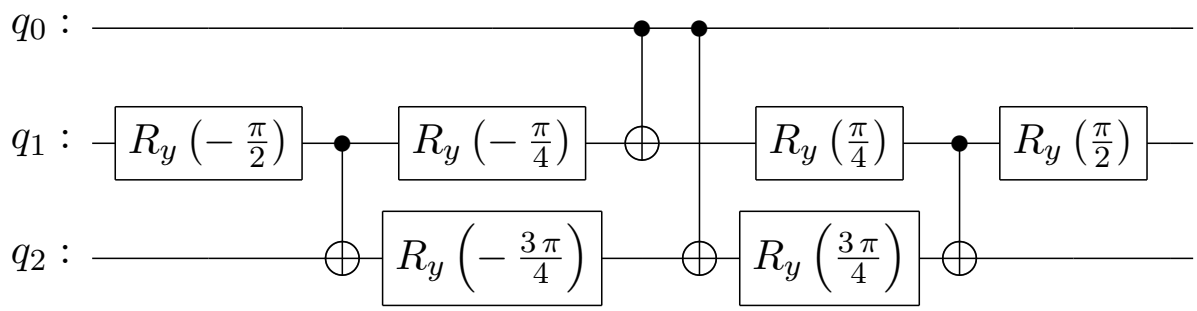


Figure 1: Literal controlled fermionic SWAP. Wire zero is c, and wires one and two are a, b. The interaction count is optimal in the Ising model.

5 An invariant characterization of the two-SQiSW region

Let $\Gamma = U(YY)U^T(YY)$ and $\Delta = \det U$. Define the scalar-independent Makhlin invariants [1]

$$g_1 = \frac{(\operatorname{tr}\Gamma)^2}{16\Delta}, \qquad g_2 = \frac{(\operatorname{tr}\Gamma)^2 - \operatorname{tr}(\Gamma^2)}{4\Delta}. \tag{7}$$

Use $\mathcal{A}(x, y, z) = \exp[i(xXX + yYY + zZZ)]$, with chamber $\pi/4 \geq x \geq y \geq |z|$, and SQiSW $= \mathcal{A}(\pi/8, \pi/8, 0)$.

Theorem 3 *A two-qubit unitary has a projective realization with at most two SQiSW pulses and arbitrary local unitaries if and only if*

$$W = g_2 + 1 \geq 0, \qquad W^2 \geq 8(|g_1| + \Re g_1). \tag{8}$$

This is an invariant rewriting of Huang et al.'s region $x \geq y + |z|$ [4], including its boundary.

Proof. In the chamber, $g_1 = (u+iv)^2$ up to the immaterial sign of v, where $u = \cos 2x \cos 2y \cos 2z \geq 0$, and $g_2 = \cos 4x + \cos 4y + \cos 4z$. Thus $|g_1| + \Re g_1 = 2u^2$, and Eq. (8) is equivalent to $W \geq 4u$. Put $X = \cos 2x$, $Y = \cos 2y$, $Z = \cos 2|z|$. Direct factorization gives

$$\begin{aligned} W - 4u &= 2(X^2 + Y^2 + Z^2 - 2XYZ - 1) \\ &= 2[X - \cos 2(y + |z|)][X - \cos 2(y - |z|)]. \end{aligned} \tag{9}$$

When $|z| > 0$, the second factor is negative: $x \geq y > y - |z|$ and cosine decreases on the relevant interval. The first factor is nonpositive exactly when $x \geq y + |z|$. When $z = 0$, the product becomes $(X - Y)^2 \geq 0$ and the chamber already satisfies $x \geq y$. Huang et al.'s necessary-and-sufficient

chamber characterization now proves both directions. Exact polynomial checks establish Eq. (9) and the identity for $(u+iv)^2$, independently of random sampling. *QED*
The test decides zero/one/two-pulse-region membership, not the exact minimum within that region. A separate zero/one-pulse class test is needed to assert that two pulses are necessary. No new universal three-pulse bound is claimed.

6 Symbolic constructions and reference resources

The implementation accepts exact matrices and named symbolic families. It enumerates all distinct nontrivial cuts, factors rank-one realignments, and peels candidate interactions in a bounded residual search. Exact exponential-polynomial arithmetic supplies rank witnesses rather than numerically thresholded SVDs. Native controlled, exchange, B, and SQiSW charts supply complementary candidates. Every accepted candidate is checked by an exact identity or a compositional certificate of proved replacements. A failed search remains unresolved, and a search deadline is not an impossibility argument.

The B and SQiSW charts include folded-angle and degenerate branches. Long expressions are stored as a scalar directed acyclic program with scoped aliases. The canonical three-SQiSW projective chart has 585 scalar bindings and 11 physical gates. This is a machine-inspectable exact expression program, not a short human-readable formula. Numerical evaluation compiles the shared dependencies once and evaluates only the branches required for a binding; the exact family verifier remains the acceptance criterion. The numerical binding experiment below measures this implementation without claiming that symbolic synthesis is always faster than numerical synthesis.

The grid contains 327 catalog entries crossed with 32 supplied alphabets. There are 217 named one-qubit entries, including 192 matrices obtained from 24 Clifford representatives and eight phases; the remaining named widths are 72 two-qubit, 14 three-qubit, three four-qubit, and one five-qubit entry. Twenty register fixtures use specified widths. Their physical widths produce 218, 79, 26, 3, and 1 entries at widths one through five. Aliases and phase multiples are included explicitly and are not independent families. In literal mode, 222 entries admit zero-entangler constructions in every preset where they are representable; zero-cost nonlocal synthesis is not inferred from the size of the headline grid.

The Clifford reference includes finite-alphabet shortest-path proofs for $\{H, S, T, T^\dagger\}$: a radius-seven search reaches 628 matrices, and meet-in-the-middle splitting proves word minima for all 192 targets, with maximum length fourteen. Continuous-alphabet circuits have no corresponding unrestricted local-optimality claim. The result is a useful explicit phase reference alongside established Clifford+T synthesis theory [15, 16, 17, 18]. The actual gate and Clifford PDFs, their editable LaTeX, alphabetical index, and exact expression glossaries are included in the full review package.

7 Evaluation

7.1 Coverage by width and proof scope

The following tables classify every effective cell, including a literal circuit reused as its projective construction. “Fixed min.” and “Family min.” are a disjoint union of matching explicit lower-bound payloads. The latter certifies a necessary uniform family budget, not the minimum at every parameter specialization. “Unproved” means that a positive-entangler construction lacks a matched certificate in this union. Legacy Boolean optimality flags alone are not counted as proof payloads. Zero-entangler constructions and obstructions appear separately.

Literal equality.

Qubits	Cells	$E = 0$	$E > 0$	Fixed min.	Family min.	Unproved	Obstructed
1	6976	5210	0	0	0	0	1766
2	2528	117	1754	689	710	355	657
3	832	80	468	43	0	425	284
4	96	0	59	0	0	59	37
5	32	0	14	0	0	14	18
Total	10464	5407	2295	732	710	853	2762

Projective equality.

Qubits	Cells	$E = 0$	$E > 0$	Fixed min.	Family min.	Unproved	Obstructed
1	6976	6915	0	0	0	0	61
2	2528	128	2247	936	955	356	153
3	832	96	692	48	0	644	44
4	96	0	89	0	0	89	7
5	32	0	29	0	0	29	3
Total	10464	7139	3057	984	955	1118	268

The row ledger lists gate, alphabet, phase policy, width, interaction count, proof scope, and every matched proof source. This resolves the earlier inconsistency between per-result flags and separate audit totals. In the pre-revision archive the raw positive-entangler flags were 694 literal and 423 explicitly stored projective results. Its 1,916 explicitly stored positive-entangler projective results omitted 1,141 positive-entangler literal circuits reused projectively. The denominator for the complete projective grid is therefore 3,057, not 1,916. Neither the flags nor an uncorrected sum of overlapping audits is our reported optimality total.

The obstruction breakdown is as follows.

Obstruction	Literal	Projective
determinant_and_clifford_t_field	0	16
determinant_group	2601	0
discrete_alphabet_nonconstant_family	142	210
not_a_clifford	13	38
outside_clifford_t_number_field	6	4

These are standard necessary conditions, used to describe feasibility. In particular, 2,601 literal obstructions are determinant-group tests, and all 268 projective obstructions occur in the three discrete Clifford or Clifford+T presets. A nonconstant continuous family cannot be covered by a countable discrete alphabet, although special parameter values can be representable. We do not present these simple feasibility checks as new lower-bound theorems.

7.2 Independent compiler comparisons

Eight fixed targets are constructed independently from library gates. Qiskit 2.6.2, pytket 2.18.5, and BQSKit 1.2.1 receive these source words; for targets on at most three qubits they also receive the target matrix. The table selects the smallest verified (E, L) pair over those recorded routes. The source is never our optimized output. Qiskit uses optimization level three and seed 219; tket uses full peephole optimization with implicit swaps disabled; BQSKit uses level three, maximum synthesis size three, and a 120-second process budget. All inputs are arbitrary states, ancilla-free,

with fixed output wires and all-to-all connectivity. Full-matrix projective replay requires maximum entry error below 10^{-8}.

Quartz and the original Java-17 QUESO implementation receive the same independent sources in the NAM alphabet $\{H, R_z, X, \mathrm{CX}\}$. Each has a 10-second search budget and optimizes total native gate count. Quartz uses its supplied NAM equivalence set; QUESO rules are generated with three wires and length six. Rule files, repository revisions, the Quartz harness patch, and output QASM are archived. Their displayed local counts use one-qubit fusion into U3, preserving the entangling skeleton; these settings do not constitute an exhaustive search for each tool's best possible result.

Gate	Qiskit	tket	BQSKit	Quartz	QUESO	Exact atlas
SYC	3/8	3/7	3/4	4/5	4/5	3/3
iSWAP	2/3	2/4	2/4	2/3	2/3	2/3
SSWAP	3/8	3/7	3/5	–	3/8	3/7
fFredkin	5/13	10/15	4/6	9/10	13/10	4/6
CCX	6/8	6/8	6/8	6/8	6/8	6/8
CSWAP	7/10	7/8	7/9	8/8	8/8	7/10
C3X	14/19	14/16	14/16	14/16	14/16	14/17
C4X	36/41	36/41	–	30/33	36/36	36/41

Entries show E/L; "–" means no output passed the declared replay threshold on the recorded route. BQSKit has one runtime error, one 120-second exhaustion, and an unverified 30-CX C4X output. Quartz's SSWAP output has error about 6.9×10^{-7} and is excluded. Every failure and its settings remain in the raw record.

BQSKit independently matches our fFredkin count of four CNOTs and six locals. A separate enumeration of all 27 three-edge and 81 four-edge layouts, with three BQSKit starts per layout, also finds a four-CNOT fit in about six seconds overall. Failed numerical three-CNOT fits are not the lower-bound proof. The eleven older SDK entangler wins were all fFredkin in different presets; they were not eleven distinct gate discoveries. The current data also contain genuine losses: Quartz reaches 30 CNOTs for C4X while this atlas entry has 36; tket and BQSKit use fewer C3X locals; BQSKit uses fewer SSWAP locals. Consequently we claim neither compiler dominance nor that all atlas cells attain the best known upper bound.

For context, the identified ancilla-free published CX constructions include 14 for C3X and 36 for C4X [19, 20], and seven for ordinary Fredkin [21]. These are published upper bounds, not general optimality theorems. Quartz's measured 30-CX C4X result is reported alongside the older 36-CX construction. Toffoli's six-CX minimum is proved [14]; the five-unrestricted-gate Fredkin result [13] uses a different primitive. UniversalQCompiler is the only listed analytical package without a run here, because its licensed Wolfram kernel was unavailable.

7.3 Component ablations

Six opaque two-qubit families are tested in B/U3 and SQiSW/U3: X/Y-controlled rotations, unequal controlled SU(2) branches, a phase-gauged exchange block, a gauged affine Cartan family, and unequal commuting exchanges. An earlier chart-disabled experiment verifies 3/12, compared with 11/12 for the full method. This establishes that native charts account for much of this workload's success.

We now separately disable OSR factor construction and residual interaction peeling, retaining the native charts, Clifford and commuting-Pauli constructors, and posthoc identity/lower-bound checks. Both methods receive opaque matrices and a 20-second outer deadline, with a 0.2-second

residual search budget where enabled. The additional three- and four-qubit probes are explicitly defined independent-product circuits, not catalog names.

Qubits	Method	Cases	Verified	Unresolved	Deadline	Errors
2	Full	12	11	0	1	0
2	OSR construction off	12	11	0	1	0
3	Full	1	0	1	0	0
3	OSR construction off	1	0	1	0	0
4	Full	1	0	1	0	0
4	OSR construction off	1	0	1	0	0

Figure 2 displays these results. OSR construction does not increase the success count on the two-qubit chart workload, and neither method resolves the wider probes within this configuration. This experiment does not support attributing the chart successes to OSR search. OSR's role in the mathematical lower bounds is established analytically, independently of the ablation. The title accordingly identifies the actual theorems.

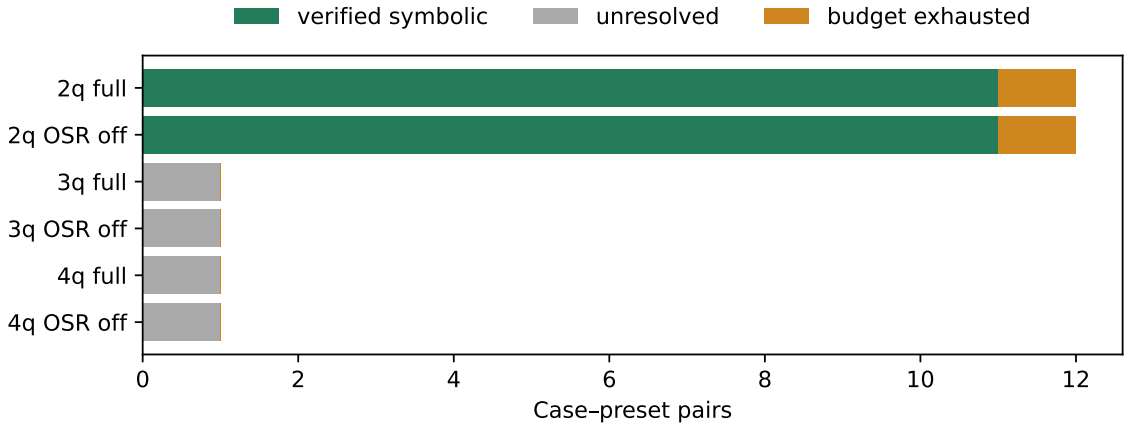


Figure 2: The full constructor and the constructor with OSR factorization and residual search disabled. The two-qubit workload is dominated by native charts. Wider unresolved probes are retained.

7.4 Parameter reuse and binding cost

The four-qubit example composes six fSim blocks on $(0,1),(2,3),(1,2)$ repeated twice, with twelve independent parameters. A gatewise source $R_{xx}(\theta)R_{yy}(\theta)CP(-\phi)$ is compared with a three-CX analytical source, using the same Qiskit 2.2.3 optimization. Three-CX fSim synthesis is established analytical knowledge, not a new resource bound.

Route	E	L	Depth	Compile (ms)	1,000 binds (ms)
Gatewise analytical source	36	66	55	50.76	114.28
Three-CX symbolic source	18	42	31	7.13	52.14
Numerical compile per binding	18	40	25	7.68	–

The numerical row is the mean compilation time over ten independently bound instances; all ten give 18/40/25. It beats the symbolic circuit in locals and depth. The weak gatewise 36/66/55 row is retained for context and is not evidence of improvement over the known three-CX construction.

Using that known analytical source as a compile-once control gives the same interaction budget as the symbolic route. Qiskit already supports parameter reuse [8]. Cold symbolic chart synthesis took 5.63 seconds in this run; the displayed SDK time excludes that cost. Ten complete 16-by-16 numerical replays check the composed family, whose constituent identities are separately proved exactly.

For the larger 585-binding scalar DAG, compilation took about 1.36 seconds and 1,000 warm evaluations about 0.97 seconds. Full two-qubit matrix replay passed for 64 parameter assignments, including the identity, an Ising corner, the SWAP-class corner, and 60 seeded points spanning multiple angle folds. These are numerical regression tests of an exactly certified chart. They show that a large symbolic program can be reused without repeated symbolic substitution; they do not establish a speed advantage over bind-time KAK on arbitrary workloads. Symbolic outputs are valuable when a free-parameter identity, phase contract, or reusable expression is needed; specialization and numerical optimization remain useful alternatives.

7.5 A benchmark-suite peephole check

We apply a deliberately simple atlas peephole pass to every existing IBMEagle and QASMBenchmarks QASM source with two to eight declared qubits and fewer than 500 semicolons, yielding eight circuits. Numerical fingerprints propose contiguous windows of at most eight operations on two wires; exact identities accept matches. The control is gatewise atlas lowering into B/U3 or SQiSW/U3, with the same lowering on unmatched gates. Every substitution is therefore compositionally exact. The materialized native words also pass 32 seeded statevector replays. Source hashes, selection rules, counts, and attempted-match timing are archived.

Circuit	Basis	Qubits	E before	L before	E after	L after	Rules
QFT8	B	8	112	316	112	316	0
QFT8	SQiSW	8	112	316	112	316	0
barenco_tof_3	B	5	48	146	48	146	0
barenco_tof_3	SQiSW	5	48	146	48	146	0
barenco_tof_4	B	7	96	290	96	290	0
barenco_tof_4	SQiSW	7	96	290	96	290	0
mod5_4	B	5	56	163	56	163	0
mod5_4	SQiSW	5	56	163	56	163	0
tof_3	B	5	36	111	36	111	0
tof_3	SQiSW	5	36	111	36	111	0
tof_4	B	7	60	185	60	185	0
tof_4	SQiSW	7	60	185	60	185	0
adder_n4	B	4	20	53	20	53	0
adder_n4	SQiSW	4	20	53	20	53	0
tfim	B	4	120	310	120	310	0
tfim	SQiSW	4	120	310	120	310	0

The pass finds no improving windows on this suite. A separate positive control replaces the standard three-CX SWAP word by its archived native construction and verifies the expected interaction reduction. This checks the exact SWAP identity and native resource accounting, but does not turn the negative suite result into an application speedup. Wider windows, commutation search, and richer parameterized matching are future optimizer work. The present evidence supports the reference library and theory, not an end-to-end peephole performance claim.

8 Artifacts and limitations

The repository is https://github.com/rakytap/sequential-quantum-gate-decomposer. The review bundle contains the manuscript, its LaTeX, both complete atlas PDFs and editable sources, theorem component checks, a row-resolved optimality ledger, full comparison circuits and failures, rules, versions, and content hashes. A scientific archive view removes coordination metadata and separates current constructions from historical attempts; original immutable archives remain in the repository. Exact source-content hashes and Git revisions identify implementations even where an exploratory run preceded its source commit. No archival DOI is claimed before deposition.

The classification concerns balanced involutions and arbitrary local unitaries, not every controlled gate or every restricted local alphabet. The Ising lower bound is broader than a fixed primitive, but does not cover arbitrary two-qubit interactions. General continuous local-cost minima, all-grid entangler optimality, and comparison with every publication remain open. Twenty retained shared-expression scores are unevaluated. Finite-alphabet shortest paths do not prove continuous-alphabet minima. The finite experiments have fixed budgets and include failures and losses; we infer no pulse latency or fidelity gain from abstract local-box counts. The new normal-form classification and expanded lower proof are analytical arguments with exact checked components, rather than proof-assistant formalizations, and warrant independent mathematical review.

Use of AI and authorship. An OpenAI Codex assistant supported implementation, calculations, experiments, source research, and drafting. The named authors are Gregory Morse and Peter Rakyta; affiliations and individual contribution statements will be supplied by the authors for submission. The authors remain responsible for checking the mathematics, software, data, and final text.

References

[1] Y. Makhlin, Nonlocal properties of two-qubit gates and mixed states and optimization of quantum computations, Quantum Information Processing 1, 243–252 (2002), arXiv:quant-ph/0002045.

[2] V. V. Shende, S. S. Bullock, and I. L. Markov, Recognizing small-circuit structure in two-qubit operators and timing Hamiltonians to compute controlled-NOT gates, Physical Review A 70, 012310 (2004), doi:10.1103/PhysRevA.70.012310.

[3] J. Zhang, J. Vala, S. Sastry, and K. B. Whaley, Minimum construction of two-qubit quantum operations, Physical Review Letters 93, 020502 (2004), arXiv:quant-ph/0312193.

[4] C. Huang et al., Quantum instruction set design for performance, Physical Review Letters 130, 070601 (2023), arXiv:2105.06074.

[5] M. Xu, Z. Li, O. Padon, S. Lin, J. Pointing, A. Hirth, H. Ma, J. Palsberg, A. Aiken, U. A. Acar, and Z. Jia, Quartz: Superoptimization of quantum circuits, Proceedings of PLDI (2022), doi:10.1145/3519939.3523433.

[6] A. Xu, A. Molavi, L. Pick, S. Tannu, and A. Albarghouthi, Synthesizing quantum-circuit optimizers, Proceedings of the ACM on Programming Languages 7, PLDI, Article 140 (2023), doi:10.1145/3591254.

[7] R. Iten, O. Reardon-Smith, E. Malvetti, L. Mondada, G. Pauvert, E. Redmond, R. S. Kohli, and R. Colbeck, Introduction to UniversalQCompiler, arXiv:1904.01072 (2019; revised 2021).

[8] IBM Quantum, Qiskit Parameter documentation, https://quantum.cloud.ibm.com/docs/en/api/qiskit/qiskit.circuit.Parameter, accessed 7 November 2026; executed versions are in the experiment manifest.

[9] D. C. McKay, C. J. Wood, S. Sheldon, J. M. Chow, and J. M. Gambetta, Efficient Z gates for quantum computing, Physical Review A 96, 022330 (2017), arXiv:1612.00858.

[10] G. Morse, M. Al Farizi, J. Nádori, L. Hajas, T. Kozsik, and P. Rakyta, Entanglement-aware quantum circuit optimization through globally optimal partitioning, arXiv:2611.03845 (2026).

[11] M. A. Nielsen, C. M. Dawson, J. L. Dodd, A. Gilchrist, D. Mortimer, T. J. Osborne, M. J. Bremner, A. W. Harrow, and A. Hines, Quantum dynamics as a physical resource, Physical Review A 67, 052301 (2003), arXiv:quant-ph/0208077.

[12] N. Yu, R. Duan, and M. Ying, Five two-qubit gates are necessary for implementing Toffoli gate, Physical Review A 88, 010304(R) (2013), arXiv:1301.3372.

[13] N. Yu and M. Ying, Optimal simulation of three-qubit gates (2013), arXiv:1301.3727.

[14] V. V. Shende and I. L. Markov, On the CNOT-cost of TOFFOLI gates, Quantum Information and Computation 9, 461–486 (2009), arXiv:0803.2316.

[15] K. Matsumoto and K. Amano, Representation of quantum circuits with Clifford and $\pi/8$ gates (2008), arXiv:0806.3834.

[16] V. Kliuchnikov, D. Maslov, and M. Mosca, Fast and efficient exact synthesis of single-qubit unitaries generated by Clifford and T gates, Quantum Information and Computation 13, 607–630 (2013), arXiv:1206.5236.

[17] B. Giles and P. Selinger, Exact synthesis of multiqubit Clifford+T circuits, Physical Review A 87, 032332 (2013), arXiv:1212.0506.

[18] B. Giles and P. Selinger, Remarks on Matsumoto and Amano's normal form for single-qubit Clifford+T operators (2013), arXiv:1312.6584.

[19] D. Maslov, Advantages of using relative-phase Toffoli gates with an application to multiple control Toffoli optimization, Physical Review A 93, 022311 (2016), arXiv:1508.03273.

[20] IBM Quantum, Qiskit synthesis API, https://quantum.cloud.ibm.com/docs/en/api/qiskit/synthesis, accessed 7 November 2026.

[21] P. Cruz and B. Murta, Shallow unitary decompositions of quantum Fredkin and Toffoli gates for connectivity-aware equivalent circuit averaging, APL Quantum 1, 016105 (2024), arXiv:2305.18128.

A Secondary objective and interpretation

The implementation orders candidate resources as (E, S, L), where E is native entanglers, L is local gates, and the optional description score is

$$S(C) = \sum_g \left[2^{k_g} + \frac{1}{5} \sum_{p \in \text{params}(g)} \left(\log_2 \max\{1, |\operatorname{den}(\text{simplify } p)|\} + \frac{1}{100} \right) \right]. \tag{10}$$

This heuristic prefers shorter parameter descriptions and named gates over unnecessary Euler parameters. It depends on the simplifier and has no calibrated pulse-error interpretation. The theorems optimize E alone. Our compiler table compares (E, L) explicitly, independently of this score; more locals can occasionally score better under the archive's objective. Unknown equal-E scores remain incomparable, not zero. A historical unclamped denominator penalty admitted negative-cost identity cycles; the nonnegative version removes that pathology. That repair is software history, not a scientific optimality result. The preserved migration audit checked that 1,390 retained angle expressions kept their scores.

B Preset definitions

Preset	Entangler	Local alphabet
cx_u	CX	U3
cx_rz_ry	CX	Rz, Ry
cx_h_phase	CX	H, P
ibm_cx	CX	Rz, Sx, X
ibm_ecr	ECR	Rz, Sx, X
ibm_cz	CZ	Rz, Sx, X
cz_u	CZ	U3
cz_rz_rx	CZ	Rz, Rx
iswap_u	iSWAP	U3
sqrt_iswap_u	SiSWAP	U3
sqrt_swap_u	SSWAP	U3
xx90_u	XX90	U3
xx_rz_ry	Rxx	Rz, Ry
zz_u	Rzz	U3
zx_u	Rzx	U3
google_cz	CZ	PhasedXZ
google_sqrt_iswap	SiSWAP	PhasedXZ
google_syc	SYC	PhasedXZ
google_fsim	fSim	PhasedXZ
berkeley_b	B	U3
ionq_aria	IonQMS	GPi, GPi2
ionq_aria_full	IonQMSMax	GPi, GPi2
ionq_forte	IonQZZ	GPi, GPi2
quantinuum_zzmax	ZZMax	PhasedX, Rz
quantinuum_zzphase	ZZPhase	PhasedX, Rz
tket_tk2	TK2	PhasedX, Rz
rigetti_cz	CZ	Rz, Sx, X
aqt_rxx	Rxx	R, Rz
iqm_cz	CZ	R
clifford_t_cx	CX	H, S, T, Tdg
clifford_t_cz	CZ	H, S, T, Tdg
clifford_only	CX	H, S

The presets specify mathematical gate alphabets, not calibrated processors. A TK2 macro is one supplied primitive, and fixed discrete local alphabets are preserved even when they obstruct a continuous target family.

C Reproduction

The source bundle includes the pinned environment manifests, current scripts, raw outputs, exact proof programs, and build inputs. The primary commands are:

```
python3 -m pytest tests/symbolic -q
python3 examples/decomposition/symbolic_referee_components.py \
 --output /tmp/components.json.gz
python3 examples/decomposition/symbolic_referee_peephole.py
python3 examples/decomposition/symbolic_referee_sdks.py \
 --output /tmp/sdks.json.gz
python3 examples/decomposition/symbolic_referee_rules.py \
 --output /tmp/rules.json.gz
```

SDK comparisons use the separate Python-3.12 environment. Rule optimizers require the recorded Quartz build and Java-17 QUESO artifact; complete rule files, commands, and the harness patch are included. The builder regenerates coverage and comparison tables from the frozen records. Compilation of the paper needs `manuscript.tex` and `generated/`; the atlas sources retain their own relative `circuits/` and `programs/` paths.